

D. Coprime Sums

time limit per test: 1 second
memory limit per test: 256 megabytes

You are given a list $a = (a_1, \dots, a_n)$ of n non-negative integers, and you must answer p queries of the form: given k , print the sum of values of a at positions that are coprime to k . In other words, return the sum of values of a_i for those i such that $\gcd(k, i) = 1$, where $\gcd(k, i)$ is the greatest common divisor of k and i .

Input

The first line of the input contains the number of cases T .

Each case starts with a line of input with two integers n and p , the size of a and the number of queries.

The following line of each case contains n integers $a_1, \dots, a_n$.

The next line of each case contains p integers, the values of k for each of the queries.

Output

For each case, you must print a line with p integers, the answers to each of the queries.

Scoring

- (13 points) The sum of n and the sum of p across all cases are at most 10^3 .
- (24 points) It is guaranteed that a_i will be non-zero only if i is prime.
- (35 points) $k \leq 1000$, $T \leq 10$, $a_i = 1$ for all i .
- (28 points) No additional restrictions.

Example

input	Copy
3 4 4 1 1 1 1 1 2 3 4 10 4 3 9 2 3 4 5 5 0 2 1 7 1 3 2 8 1 1000000000 1000000000 1000000000 1000000000 1000000000 1000000000 1000000000 1000000000 1	
output	Copy
4 2 3 2 29 34 25 16 8000000000	

Note

- $1 \leq T \leq 10^5$.
- $1 \leq n, p \leq 2 \cdot 10^5$. The sum of n and the sum of p across all cases are at most $2 \cdot 10^5$.
- $0 \leq a_i \leq 10^9$.
- $k \leq n$.

XXIX Spain Olympiad in Informatics, Online Qualifier

Finished

Practice

About Contest

Contest website

Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

Submit?

Language: GNU G++20 13.2 (64 bit, win

Choose file: Choose File No file chosen

Submit

Last submissions

Submission	Time	Verdict
330413145	Jul/22/2025 20:51	Perfect result: 100 points
330412765	Jul/22/2025 20:48	Partial result: 0 points
323064644	Jun/06/2025 05:11	Perfect result: 100 points
310953703	Mar/17/2025 00:21	Perfect result: 100 points
310951784	Mar/16/2025 23:57	Perfect result: 100 points
310950291	Mar/16/2025 23:40	Perfect result: 100 points
310949756	Mar/16/2025 23:34	Partial result: 13 points
310947430	Mar/16/2025 23:10	Partial result: 13 points
310946846	Mar/16/2025 23:04	Partial result: 13 points
310946427	Mar/16/2025 23:00	Partial result: 0 points